

SOFTWARE REQUIREMENTS SPECIFICATION

AI-POWERED UPSKILLING RECOMMENDATION AND AUTOMATION SYSTEM

Project ID: 22

Course: Software Engineering Mini-Project

Department: Computer Science and Engineering (AI & ML)

University: PES University, Bengaluru

SUBMITTED BY

Sl. No.	Name	SRN
1	POOJITHA V	PES2UG24AM113
2	MANASA C	PES2UG25AM803
3	SHANMUGA	PES2UG24AM083
4	MANISH S	PES2UG24AM085

Faculty: Dr. Siba Ram Baral

Academic Year: 2026–2027

REVISION HISTORY

Version	Date	Description
1.0	05-09-2026	Initial SRS document

1. INTRODUCTION

1.1 Purpose

The purpose of this project is to develop an **AI-Powered Upskilling Recommendation and Automation System** that helps students identify the skills required for their desired career roles and recommends suitable learning resources.

The system analyzes the student's existing skills, courses, projects, certifications, and career goals to identify skill gaps and generate a personalized learning path.

1.2 Intended Audience

This Software Requirements Specification is intended for students, developers, testers, project guides, and administrators involved in the development and use of the system.

1.3 Product Scope

The system provides personalized upskilling guidance to students. It allows students to maintain their profile, enter their skills and qualifications, select a career goal, identify skill gaps, receive AI-based recommendations, and track their learning progress.

The main objective is to help students understand **what skills they are missing and what they should learn next** to achieve their desired career goal.

1.4 References

- JACKFRUIT Software Engineering Mini-Project Guidelines 2026
- PES University Software Engineering Project Guidelines

2. OVERALL DESCRIPTION

2.1 Product Perspective

The proposed system is a web-based application that acts as a personalized career and learning assistant.

The student provides information about their current skills and career goal. The system compares the student's skills with the skills required for the selected career role and generates suitable recommendations.

System Flow

Student → Profile → Skills → Career Goal → Skill Gap Analysis → AI Recommendations → Learning Path → Progress Tracking

2.2 Product Functions

The main functions of the system are:

- User registration and login
- Student profile management
- Skill management
- Course management
- Project management
- Certification management
- Career goal selection
- Skill gap analysis
- AI-powered recommendations
- Learning resource recommendations
- Learning progress tracking
- Dashboard
- Administrator management

2.3 User Classes

Student

The student is the primary user. Students can manage their profile, enter their skills, select career goals, view skill gaps, receive recommendations, and track their progress.

Administrator

The administrator manages student information, skills, career roles, courses, and learning resources.

2.4 Operating Environment

The system will operate as a web application.

Hardware:

- Laptop/Desktop/Smartphone
- Internet connection

Software:

- Web browser
- Web server
- Database
- AI/ML recommendation module

2.5 Constraints

- Internet connectivity is required.
- Student information must be securely stored.
- Recommendations depend on the available data.
- The system should have a simple and user-friendly interface.
- The AI recommendation module requires appropriate training or reference data.

2.6 Assumptions and Dependencies

- Students provide correct information about their skills and qualifications.
- Career-role requirements are available in the system.
- Suitable learning resources are available.
- The AI recommendation module can access the required student and career data.

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 User Interface

The system will provide the following interfaces:

- Registration page
- Login page
- Student profile page
- Skills page
- Career goal page

- Skill gap analysis page
- Recommendation page
- Learning progress page
- Dashboard
- Administrator panel

The interface should be simple, clear, and easy to navigate.

3.2 Software Interfaces

The system will interact with:

- Database management system
- AI/ML recommendation module
- Authentication module
- Web server
- Learning-resource database

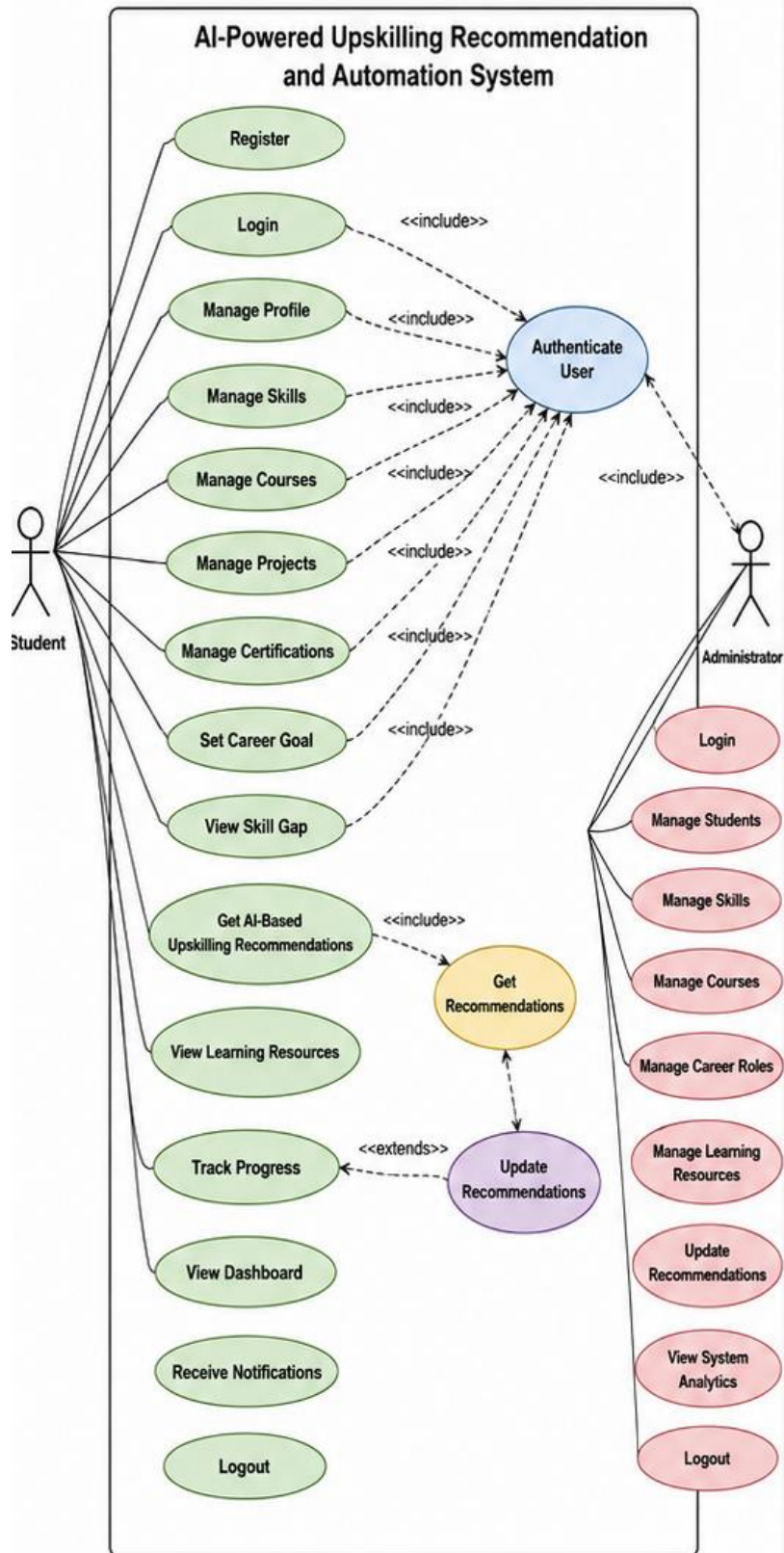
3.3 Communication Interface

The system will use standard web communication protocols such as HTTP/HTTPS for communication between the user interface, server, and database.

4. ANALYSIS MODELS

4.1 USE CASE DIAGRAM

USE CASE DIAGRAM



Actors

Student

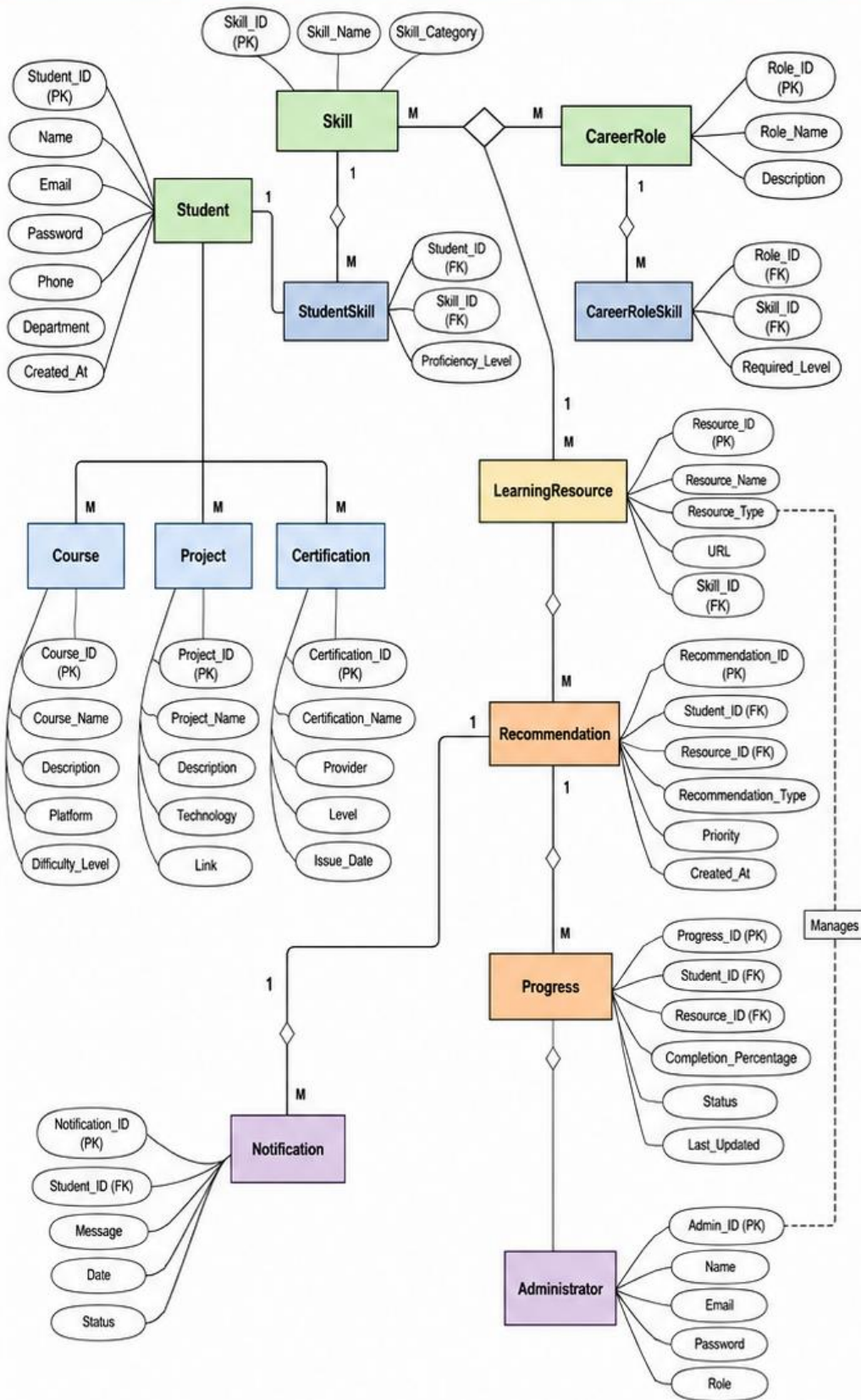
- Register
- Login
- Manage Profile
- Manage Skills
- Add Courses
- Add Projects
- Add Certifications
- Set Career Goal
- View Skill Gap
- Get AI Recommendations
- View Learning Resources
- Track Progress
- View Dashboard

Administrator

- Login
- Manage Students
- Manage Skills
- Manage Career Roles
- Manage Courses
- Manage Learning Resources

4.2 ENTITY RELATIONSHIP DIAGRAM

ER DIAGRAM



Main Entities

- Student
- Skill
- Career Role
- Course
- Project
- Certification
- Learning Resource
- Recommendation
- Progress

Relationships

- A student can have multiple skills.
- A student can complete multiple courses.
- A student can have multiple projects.
- A student can have multiple certifications.
- A student can select a career role.
- A career role can require multiple skills.
- The system generates recommendations for students.
- Students can track their progress for recommended learning resources.

5. SYSTEM FEATURES

5.1 User Registration and Login

The system shall allow students to register and securely log in.

FR-01: The system shall allow a new student to register.

FR-02: The system shall validate registration information.

FR-03: The system shall allow registered students to log in.

FR-04: The system shall display an error message for invalid login credentials.

5.2 Profile Management

The system shall allow students to maintain their personal and academic information.

FR-05: The system shall allow students to create a profile.

FR-06: The system shall allow students to edit their profile.

FR-07: The system shall store student academic and career information.

5.3 Skill Management

The system shall allow students to maintain their existing skills.

FR-08: The system shall allow students to add skills.

FR-09: The system shall allow students to update or remove skills.

FR-10: The system shall store student skill information.

5.4 Career Goal Management

The system shall allow students to select their desired career role.

FR-11: The system shall display available career roles.

FR-12: The student shall be able to select a career goal.

FR-13: The system shall store the selected career goal.

5.5 Skill Gap Analysis

The system shall compare the student's existing skills with the skills required for the selected career role.

FR-14: The system shall identify the student's existing skills.

FR-15: The system shall identify skills required for the selected career role.

FR-16: The system shall identify missing or weak skills.

FR-17: The system shall display the identified skill gaps.

5.6 AI-Powered Recommendation

The system shall generate personalized recommendations based on the student's skill gaps and career goal.

FR-18: The system shall analyze the student's skill gaps.

FR-19: The system shall generate personalized recommendations.

FR-20: The system shall recommend relevant courses and learning resources.

FR-21: The system shall recommend suitable projects.

FR-22: The system shall recommend relevant certifications where applicable.

5.7 Learning Resources

The system shall provide learning resources for improving identified skills.

FR-23: The system shall display recommended learning resources.

FR-24: The system shall categorize learning resources according to skills.

FR-25: The system shall allow students to view recommended resources.

5.8 Progress Tracking

The system shall allow students to monitor their learning progress.

FR-26: The system shall allow students to mark learning activities as completed.

FR-27: The system shall calculate learning progress.

FR-28: The system shall display learning progress on the dashboard.

FR-29: The system shall update recommendations based on learning progress where applicable.

5.9 Dashboard

The dashboard shall provide an overview of the student's learning journey.

FR-30: The system shall display the student's career goal.

FR-31: The system shall display identified skill gaps.

FR-32: The system shall display recommended learning resources.

FR-33: The system shall display learning progress.

5.10 Administrator

The administrator shall manage the data required by the system.

FR-34: The administrator shall manage student information.

FR-35: The administrator shall manage skills.

FR-36: The administrator shall manage career roles.

FR-37: The administrator shall manage courses and learning resources.

FR-38: The administrator shall update career-role skill requirements.

6. NON-FUNCTIONAL REQUIREMENTS

6.1 Performance

NFR-01: The system should respond to normal user requests within a reasonable time.

NFR-02: Recommendation results should be generated without unnecessary delay.

6.2 Security

NFR-03: User passwords shall be stored securely.

NFR-04: Only authenticated users shall access their personal information.

NFR-05: Administrator functions shall be accessible only to authorized administrators.

6.3 Usability

NFR-06: The system shall provide a simple and user-friendly interface.

NFR-07: Navigation between different pages shall be clear and consistent.

6.4 Reliability

NFR-08: The system should provide recommendations based on the available data.

NFR-09: Student progress should not be lost during normal system operation.

6.5 Maintainability

NFR-10: The system should use modular components so that individual features can be updated easily.

7. DATABASE REQUIREMENTS

The system shall maintain information related to the following entities:

Entity	Main Information
Student	Student ID, name, email, password
Skill	Skill ID, skill name
Career Role	Role ID, role name, required skills
Course	Course ID, course name, skill
Project	Project ID, project name, required skill
Certification	Certification ID, certification name
Learning Resource	Resource ID, title, link, skill
Recommendation	Recommendation ID, student, resource
Progress	Progress ID, student, resource, completion status

8. BUSINESS RULES

1. A student must register before accessing personalized features.
2. A student should provide a career goal to receive career-specific recommendations.

3. Recommendations should be based on the student's existing skills and identified skill gaps.
4. Only authorized administrators can modify system-level career and learning-resource information.
5. Student learning progress should be recorded for future recommendations.

9. REQUIREMENT TRACEABILITY MATRIX

Requirement System Feature

FR-01 – FR-04 Registration and Login

FR-05 – FR-07 Profile Management

FR-08 – FR-10 Skill Management

FR-11 – FR-13 Career Goal Management

FR-14 – FR-17 Skill Gap Analysis

FR-18 – FR-22 AI Recommendation

FR-23 – FR-25 Learning Resources

FR-26 – FR-29 Progress Tracking

FR-30 – FR-33 Dashboard

FR-34 – FR-38 Administrator

10. GLOSSARY

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
SRS	Software Requirements Specification
Skill Gap	Difference between existing and required skills

Career Role	Desired professional or job role
Recommendation	Learning suggestion provided by the system
Learning Path	A sequence of learning activities
Administrator	Person responsible for managing the system

11. CONCLUSION

The **AI-Powered Upskilling Recommendation and Automation System** is designed to help students make informed decisions about their career development.

The system analyzes the student's skills, qualifications, and career goals to identify skill gaps and provide personalized recommendations for courses, projects, certifications, and other learning resources.

By tracking learning progress and updating recommendations, the system can provide students with a structured and personalized approach to upskilling.